

Violin Plot Analysis: Exercise & Diet Pulse Rates

Seaborn Visualization Analysis & Summary Report

Visualization Overview

This report analyzes the violin plot displaying heart rate (**pulse**) distributions across exercise durations (**1 min**, **15 min**, **30 min**), activity types (**rest**, **walking**, **running**), and diet conditions (**no fat** vs **low fat**).

Summary Table of Median Pulse Rates

Diet Condition	Duration	Rest (Blue)	Walking (Orange)	Running (Green)
no fat	1 min	~92 bpm	~95 bpm	~98 bpm
	15 min	~93 bpm	~97 bpm	~125 bpm
	30 min	~94 bpm	~98 bpm	~141 bpm
low fat	1 min	~89 bpm	~91 bpm	~95 bpm
	15 min	~91 bpm	~94 bpm	~105 bpm
	30 min	~92 bpm	~95 bpm	~113 bpm

Key Takeaways

- **Activity Level:** Running drives a dramatic, time-dependent increase in pulse rate, whereas resting and walking remain stable (80–100 bpm) across all durations.
- **Duration Impact:** Prolonged running (30 minutes) shows the highest variance and peak heart rate elevation.
- **Dietary Influence:** Subjects on the "no fat" diet experienced significantly higher median pulse rates while running (up to ~141 bpm at 30 min) compared to those on the "low fat" diet (~113 bpm at 30 min).